Orthogonal Dimensional Navigation, Metric Signatures, and Causality Preserving Topologies in Bulk-Brane Spacetimes

Anonymous Researcher
Department of Theoretical Physics, Institute for Advanced Cosmological Studies

August 31, 2026

Abstract

We examine the theoretical implications of a hypothetical four-dimensional (4D) spatial bulk (x, y, z, w) interacting with a three-dimensional (3D) Minkowskian brane (x, y, z). Utilizing a geometric formulation analogous to Abbot's Flatland paradigm, we analyze whether orthogonal motion through a higher spatial dimension allows for macro-historical retro-causality or past time travel. We demonstrate that while a 4D spatial object can project arbitrary 3D spatial intersections (shadows) across a macroscopic plane ($L \sim 4\times10^6$ ly), temporal progression remains constrained by the invariant sign of the metric tensor component g_{00}. We reconcile this geometric framework with the Novikov Self-Consistency Principle and the Many-Worlds Interpretation (MWI), showing that a unified bulk-brane topology structurally forbids classical causal paradoxes.

1 Introduction

Classical general relativity permits local solutions with Closed Timelike Curves (CTCs), yet macroscale backward time travel remains hindered by severe causal paradoxes, most notably the grandfather paradox. Concurrently, Kaluza-Klein models and modern string theories expand the universe's manifold framework by embedding our 3D space into a higher-dimensional bulk.

A common thought experiment suggests that if an entity moves orthogonally to our 3D spatial plane into a 4D spatial bulk, it could bypass 3D spatial boundaries and re-enter our timeline at an arbitrary prior point. This paper evaluates the mathematical validity of this assertion by contrasting spatial coordinates (w) against temporal coordinates (t).

2 The Mathematical Signature of Time vs. Space

To understand why orthogonal spatial movement fails to alter temporal coordinates, we must examine the metric tensor $g_{\mu\nu}$ of the embedding manifold. In a flat $(4 + 1)$-dimensional bulk, the line element ds^2 is defined by:

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2 + dw^2 \tag{1}$$

The crucial distinction lies in the metric signature $(-, +, +, +, +)$. The temporal component carries a negative sign ($g_{00} = -1$), while all spatial dimensions, including the orthogonal bulk dimension w, carry positive signs ($g_{ww} = +1$).

Because $g_{ww} > 0$, any displacement Δw along the higher spatial axis preserves the purely spatial interval nature of the trajectory. Moving along w alters an object's spatial position in the bulk but exerts zero mathematical pull on the dt differential.

3 Geometric Shadowing and Light Cone Boundaries

When a 4D spatial object intersects our 3-brane, it projects a 3D intersection (a geometric "shadow"). As the object moves along w, this shadow can instantly manifest at any point along a macroscopic 3D coordinate space spanning a length L:

$$L \approx 4 \times 10^6 \text{ light-years} \tag{2}$$

While this allows for instantaneous non-local spatial teleportation from the perspective of a 3D observer, it does not constitute temporal displacement. The shadow's arrival coordinate is strictly bound to the concurrent cosmological time step t_{present}.

Every local event E on the brane remains rigidly bounded by its future and past light cones. The boundary of the light cone represents the propagation of light ($ds^2 = 0$). For a traveler to slip into the past, their worldline must achieve a timelike displacement where:

$$c^2\Delta t^2 > \Delta x^2 + \Delta y^2 + \Delta z^2 + \Delta w^2 \quad \text{with} \quad \Delta t < 0 \tag{3}$$

As shown in Equation 3, adding a positive Δw^2 term increases the spatial separation, making it mathematically harder, not easier, to satisfy a backward timelike trajectory.

Note: The geometric evolution of worldlines crossing light-cone thresholds can be mapped conceptually via established standard models, such as the Minkowski Light Cone Schema.

4 Causality Bounds: Reconciling the Frameworks

If we assume a hypothetical manifold where the w-axis maps to non-linear slices of the temporal dimension (t), an intersection with a prior frame t_{past} triggers an immediate disruption of the local light cone topology. To maintain structural stability, the universe must satisfy one of two primary quantum cosmological interpretations.

4.1 The Localized Self-Consistency Boundary

Under the Novikov Self-Consistency Principle, any orthogonal intersection with a past coordinate t_{past} is mathematically invariant. The actions of the 4D object's 3D shadow must match the exact boundary conditions that originally formulated the future state t_{future}. The path integral formulation for the history of the universe sum-over-histories yields a probability of zero for any paradox-inducing trajectory:

$$P(\text{paradox}) = \int \mathcal{D}[\phi]e^{iS[\phi]} = 0 \tag{4}$$

4.2 The Many-Worlds Branching Topology

Alternatively, modifying Everett's Many-Worlds Interpretation to account for higher dimensions suggests that the 4D bulk contains infinite parallel 3-brane sheets. An orthogonal descent into a past configuration does not intersect the traveler's native brane ($\mathcal{B}_A$). Instead, it creates or intersects a divergent branch ($\mathcal{B}_B$).

As detailed in Table 1, both structurally sound models guarantee that the traveler's native timeline remains perfectly insulated from retrocausal degradation.

5 Conclusion

The geometric elegance of moving orthogonally through a fourth spatial dimension provides a viable framework for macroscale spatial teleportation and non-local interactions. However, it

Table 1: Comparative Matrix of Higher-Dimensional Time Intersections

Interpretation	**Impact on Native Timeline ($\mathcal{B}_A$)**	**Causality Status**
Block Universe / Novikov	Unchanged (traveler was always there)	Preserved (Deterministic)
Many-Worlds (MWI)	Unchanged (traveler vanishes from $\mathcal{B}_A$)	Preserved (Divergent)
Classical Past Travel	Rewritten / Erased	Violated (Paradoxical)

does not bypass the fundamental thermodynamic and causal arrows of the temporal dimension. Whether the universe operates as a deterministic static block or a splitting quantum multiverse, causality remains globally protected against higher-dimensional intrusions.

References

[1] A. Einstein, *Relativity: The Special and General Theory*, Henry Holt and Co., 1920.

[2] I. D. Novikov, *Evolution of the Universe*, Cambridge University Press, 1983.

[3] H. Everett, "Relative State Formulation of Quantum Mechanics," *Reviews of Modern Physics*, vol. 29, pp. 454–462, 1957.